

Answer ALL questions. ALL working should be shown.

1. Given a rotation matrix in the equivalent angle-axis form:

$${}^A_B R = \begin{bmatrix} k_x^2 v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y^2 v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z^2 v\theta + c\theta \end{bmatrix}$$

- (a) Show that the (1,1) element of the matrix can be written in the quaternion form:

$$1 - 2\varepsilon_2^2 - 2\varepsilon_3^2$$

[5 marks]

- (b) Show that the (2,1) element of the matrix can be written in the quaternion form:

$$2(\varepsilon_1 \varepsilon_2 + \varepsilon_3 \varepsilon_4)$$

[3 marks]

- (c) When calculating the angle θ from the rotation matrix, it was determined that $\theta = 180^\circ$. As such, we cannot use the formulae in lecture notes to obtain k_x , k_y and k_z . Derive alternative expressions to calculate k_x , k_y and k_z in such a situation (i.e. when $\theta = 180^\circ$).

[5 marks]

- (d) Show that the first column of the rotation matrix is of unit length.

[2 marks]

2. A human wrist has 3 degrees of freedom, as shown in Figures 2.1, 2.2 and 2.3.

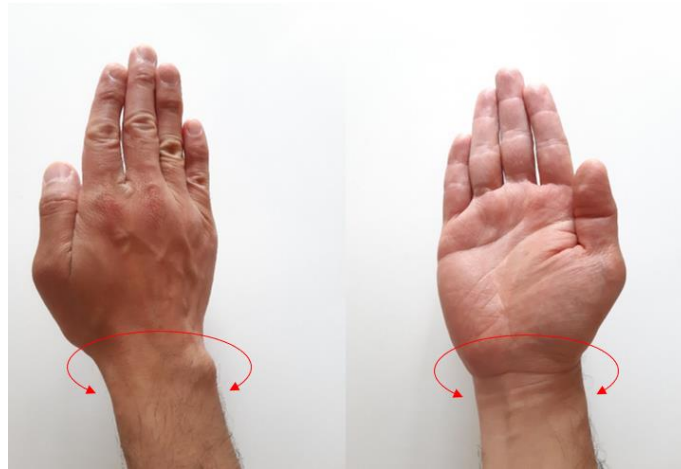


Figure 2.1: 1st degree of freedom – Pronation / Supination

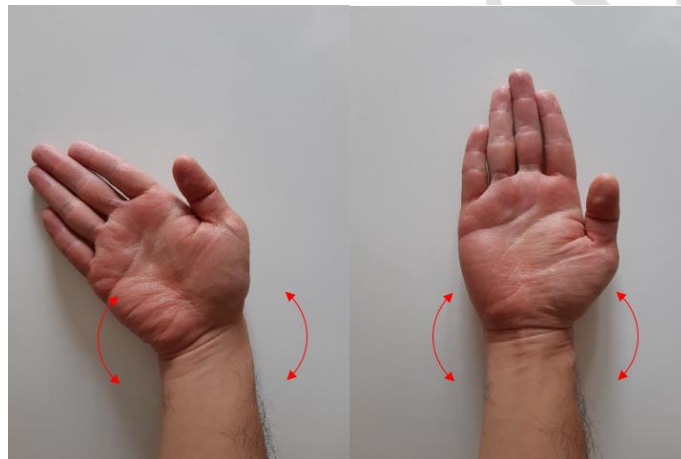


Figure 2.2: 2nd degree of freedom – Deviation

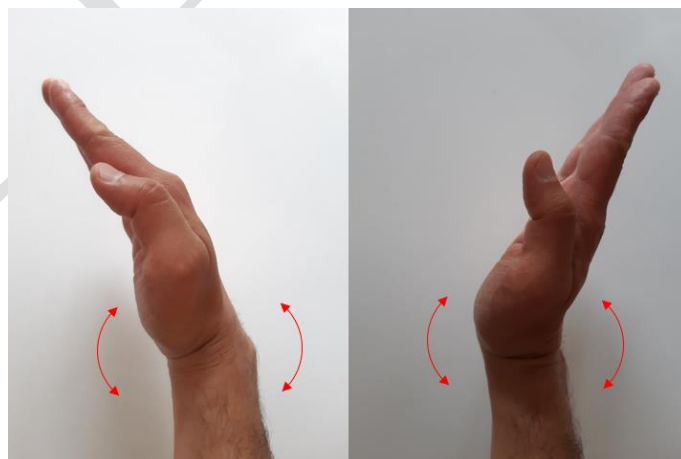


Figure 2.3: 3rd degree of freedom: Flexion / Extension

- (a) Derive the DH parameters of the wrist, where the order of the motions is first pronation / supination, followed by deviation, and ending with flexion / extension. The x_0 axis is given, and

the configuration as shown in Figure 2.4 is when $\theta_1 = 0^\circ$. Show that the transformation matrix from frame {0} to frame {3} is:

$${}^0_3T = \begin{bmatrix} c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3 & -c\theta_1 c\theta_2 s\theta_3 + s\theta_1 c\theta_3 & c\theta_1 s\theta_2 & 0 \\ s\theta_1 c\theta_2 c\theta_3 - c\theta_1 s\theta_3 & -s\theta_1 c\theta_2 s\theta_3 - c\theta_1 c\theta_3 & s\theta_1 s\theta_2 & 0 \\ s\theta_2 c\theta_3 & -s\theta_2 s\theta_3 & -c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

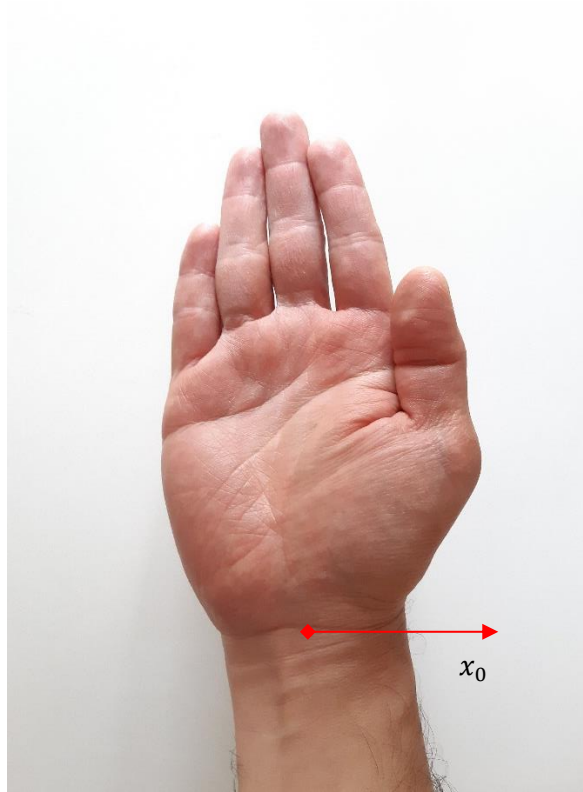


Figure 2.4: Initial configuration of wrist

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.

[8 marks]

(b) Derive the rotational Jacobian matrix of the wrist, with respect to the wrist's frame {0}. Does the wrist have any rotational singularity?

Note: If you do not have the intermediate transformation matrices in part (a), or if you are unsure of your answers in part (a), please use the following for calculation of part (b):

$${}^0_1T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta_1 & -s\theta_1 & 0 \\ 0 & s\theta_1 & c\theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} c\theta_2 & 0 & s\theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -s\theta_2 & 0 & c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[12 marks]

3. Calculate the mass moments of inertia, I_{xx} and I_{yy} (with respect to the centre of mass) of a solid long rod of length l , with an equilateral triangular cross-sectional area, as shown in Figure 3.1. The mass of the rod is m , and the density is ρ .

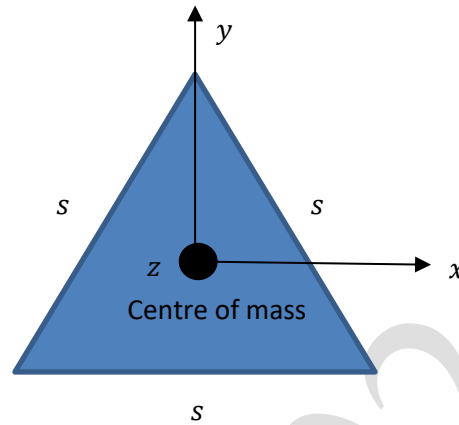


Figure 3.1: Equilateral triangle

[20 marks]

4. We want to move a robot's joint from θ_0 to θ_f in time t_f along a cubic trajectory, with zero velocities at the starting position and final position. Assume $\theta_f > \theta_0$.

(a) The values of θ_0 and θ_f are known. What should t_f be, if the velocity and acceleration cannot exceed $\dot{\theta}_{max}$ and $\ddot{\theta}_{max}$ respectively?

[10 marks]

(b) You are a creative student and decided to design your own trajectory rather than the ones you learnt in the class. One idea you have is to use the half cycle of a cosine function as a trajectory. Write down the equation of such a trajectory, parameterized by θ_0 , θ_f and t_f . Please show how you arrive at the equation, e.g. by sketches of the curves.

[10 marks]

5.

(a) For a single joint of the manipulator, the dynamics of the load is:

$$(I + I_m \eta^2) \ddot{\theta} + (b + b_m \eta^2) \dot{\theta} = \tau - \tau_{\text{fric}}$$

where $I = 10$, $I_m = 1$, $b = 5$, $b_m = 1$, $\eta = 5$, and τ_{fric} is a viscous friction with gradient $k = 0.2$.

Derive a controller such that the single joint can track a desired trajectory $(\theta_d, \dot{\theta}_d, \ddot{\theta}_d)$ with a closed-loop stiffness of 5 and closed-loop damping of 4. Sketch the implementation of the controller (e.g. sensor input, control output).

[6 marks]

(b) Assume that the controller in part (a) has been designed correctly. Determine the equation of the tracking error $e = \theta_d - \theta$, if the initial tracking error is 4, and the initial rate of change of tracking error is 0. Sketch a time-plot of the tracking error, as well as the motion of the joint if $\theta_d = 4$ (a constant).

[10 marks]

(c) Describe the behaviour of the θ response, if the closed-loop stiffness gets smaller and smaller.

[4 marks]

6. Frame $\{B\}$ is initially coincident with frame $\{A\}$. First, $\{B\}$ is rotated about z_A by β degrees, and is subsequently rotated about y_A by α degrees, followed by a rotation about z_A by γ degrees. If the resultant rotation matrix has the following numerical values:

$${}^A_B R = \begin{bmatrix} -0.3062 & -0.9186 & 0.25 \\ 0.8839 & -0.1768 & 0.4330 \\ -0.3536 & 0.3536 & 0.8660 \end{bmatrix}$$

calculate the angles α, β and γ .

[5 marks]